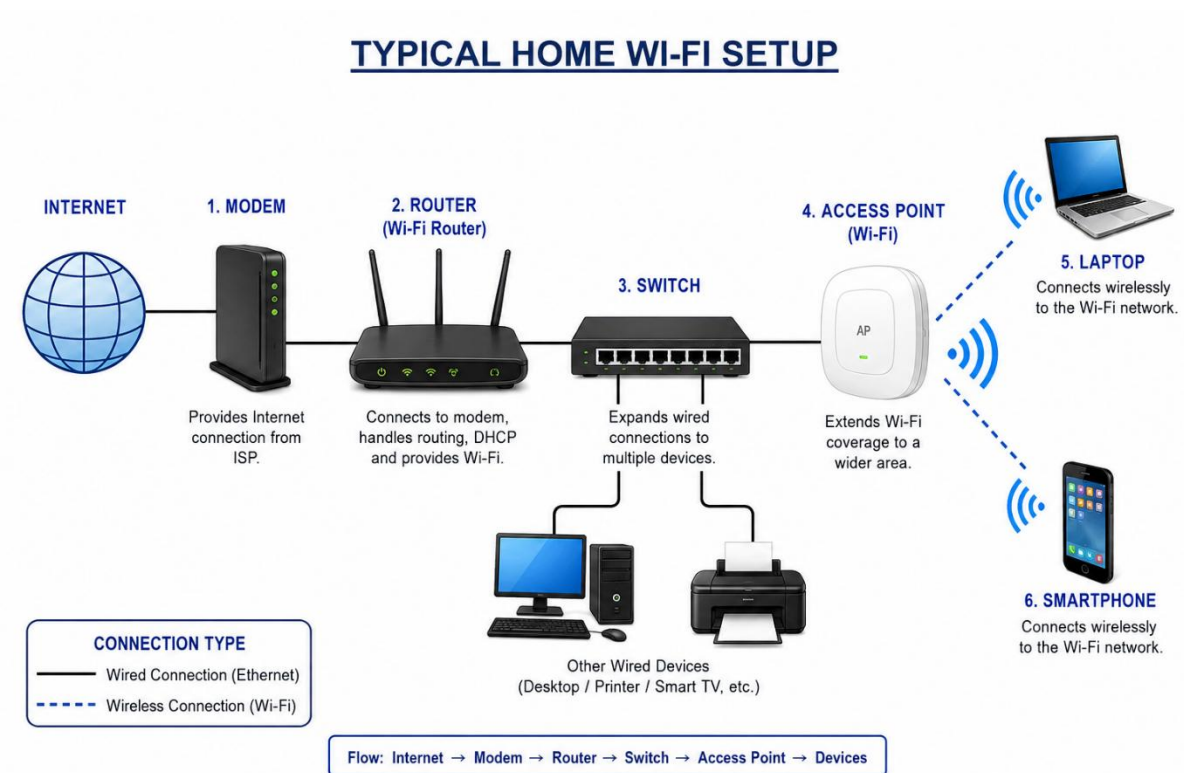


- 1) Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Internet-Modem-Router-Switch-Access point



2) List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Hardware	Main Function	Real-Life Example
Modem	Connects your home or office network to the Internet service provider (ISP) by converting signals between the ISP and your network.	At home to connect your internet service from an ISP such as Airtel or Jio.
Router	Directs network traffic between devices and connects the local network to the Internet. It can also provide Wi-Fi.	At home to share one Internet connection with laptops, smartphones, and smart TVs.
Switch	Connects multiple wired devices on the same local network and forwards data to the correct device.	In an office to connect computers, printers, servers, and other Ethernet devices.
Access Point (AP)	Provides wireless (Wi-Fi) access to a wired network, allowing wireless devices to connect.	In a school or office to provide Wi-Fi coverage to students or employees.
Network Cable	Carries data between wired networking devices, such as computers, switches, routers, and access points.	At home or in an office using an Ethernet cable to connect a PC to a router or switch.

3) Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly use



1. Ethernet Cable (RJ45 / Cat5e / Cat6)

Common use: Used to connect computers, routers switches, printers, and other network devices in a wired LAN.

Source: https://www.cisco.com/c/en/us/support/docs/routers/7000-series-routers/12223-14.html?utm_source=chatgpt.com

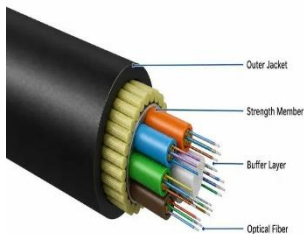
2. Coaxial Cable



Common use: Commonly used for cable TV and broadband internet connections between the service provider and homes/businesses.

Source: https://www.techtarget.com/searchnetworking/definition/coaxial-cable-illustrated?utm_source=chatgpt.com

3. Fiber Optic Cable



Common use: Used for high-speed internet and long-distance communication, including telecom networks and data centers. Fiber carries data using pulses of light.

Source: https://www.thefoa.org/tech/ref/?utm_source=chatgpt.com

4) Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Router	Switch
Connects different networks	Connects devices in one network
Uses IP addresses	Uses MAC addresses
Usually connects to the Internet	Usually connects local device
Example: Home Wi-Fi router	Example: Office network switch